

DESIGN BASIS DOCUMENT

GREENFIELD CONVENTIONAL AMMONIA PLANT

1. Purpose

The purpose of this project is to develop a **conceptual process design of a greenfield conventional ammonia plant**, covering the process simulation, material and energy balance, major equipment design basis, PFD, conceptual P&ID, utilities, process control and preliminary process-safety review. The design is based on the current DWSIM simulation and supporting project calculations.

2. Scope

The project covers:

- Natural gas feed preparation
- Air and steam feed
- Primary reforming
- Shift conversion
- CO₂ removal
- Methanation
- Synthesis-gas compression
- Ammonia synthesis
- Ammonia cooling and separation
- Liquid ammonia product
- Off-gas
- Process heat recovery
- Major equipment process design
- PFD
- Conceptual P&ID
- Control philosophy
- Preliminary PSV/relief review
- Preliminary HAZOP
- Utility summary

The project does **not** cover detailed mechanical, civil, structural, electrical or construction engineering.

3. Plant Design Basis

Parameter	Design Basis
Plant type	Conventional ammonia plant
Design level	Conceptual / preliminary process design
Simulation software	DWSIM 10.2
Simulation date	02-10-2026
Simulation basis	Lean final-basis / simplified single-pass
Operating basis	Steady-state
Main product	Liquid ammonia

4. Plant Capacity

Parameter	Value
Liquid NH ₃ production	5,453.12 kg/h
Liquid NH ₃ production	5.453 t/h
Liquid NH ₃ production	130.875 t/day
NH ₃ purity	95.4193 mol%

5. Feed Basis

Feed	Tag	Flow	Pressure
Natural Gas	NG-101	30,000 kg/h	40 bar(a)
Air	AIR-101	45,000 kg/h	40 bar(a)
Steam	STM-101	87,210 kg/h	40 bar(a)
Total	—	162,210 kg/h	—

6. Product Basis

Parameter	Value
Product stream	NH3-LIQ-1201
Product phase	Liquid
NH ₃ flow	5,453.12 kg/h
NH ₃ production	5.453 t/h
NH ₃ production	130.875 t/day
NH ₃ purity	95.4193 mol%
Temperature	30°C
Pressure	145 bar(a)

7. Process Configuration

The plant process is arranged in the following sequence:

Natural Gas + Air + Steam

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Feed Preparation

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Primary Reforming

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High-Temperature Shift

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Low-Temperature Shift

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CO₂ Removal

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Methanation

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Synthesis Gas Compression

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Ammonia Synthesis

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Cooling / Condensation

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Vapor-Liquid Separation

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Liquid NH₃ Product

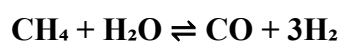
8. Key Operating Conditions

Location	Temperature	Pressure
Natural Gas Feed	30C	40 bar(a)
Air Feed	25C	40 bar(a)
Steam Feed	400C	40 bar(a)
Methanation Outlet	40°C	—
NH ₃ Synthesis Reactor Inlet	430°C	149 bar(a)
Raw Synthesis Gas after Reactor	430°C	147 bar(a)
Final NH ₃ Product	30°C	145 bar(a)

9. Reaction Basis

9.1 Primary Reforming

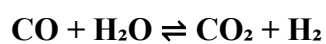
Main reaction:



The reforming section converts hydrocarbon feed into synthesis gas.

9.2 Shift Conversion

Main reaction:



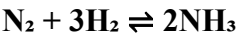
The shift section increases hydrogen production and converts CO to CO₂.

9.3 Methanation

Residual CO/CO₂ are converted to methane before ammonia synthesis.

9.4 Ammonia Synthesis

Main reaction:



The ammonia synthesis reactor operates at approximately **149 bar(a) and 430°C at the inlet** in the current simulation basis. 05_Final_Plant_Summary

10. Simulation Basis

Item	Basis
Software	DWSIM 10.2
Simulation file	Sadiq Conventional Ammonia Plant.dwxmz
Simulation date	02-10-2026
Calculation type	Steady-state
Process basis	Simplified single-pass
Thermodynamics	Peng Robinson
Components	As defined in final DWSIM model

11. Energy Design Basis

Parameter	Value
Total heating duty	74.134 MW
Total cooling duty	136.381 MW
Internal heat recovery	38.743 MW
Compressor power	0.817 MW

12. Utility Basis

Utility	Current Status
Heating duty	74.134 MW
Cooling duty	136.381 MW
Compressor electrical load	0.817 MW
Internal heat recovery	38.743 MW

13. Equipment Design Basis

Equipment	Main Design Basis
K-101	Gas flow, suction/discharge P, T, pressure ratio, power
R-101	Feed/outlet conditions, reaction, pressure drop, conversion/extent
V-101	Gas/liquid flow, P, T, vapor-liquid separation
P-101	Liquid NH ₃ flow, suction pressure, discharge pressure, head
Heaters	Feed conditions and heat duty
Coolers	Feed/outlet conditions and cooling duty
Heat exchangers	Process-side conditions and heat recovery duty

16. Battery Limits

Battery-In

- Natural gas
- Air
- Steam
- Required utilities

Battery-Out

- Liquid ammonia product
- Final off-gas
- CO₂ removal outlet
- Required utility returns/discharges

The current final stream summary identifies the liquid NH₃ product, final off-gas and CO₂-removal outlet as major final outlet streams. 05_Final_Plant_Summary

17. Design Basis Freeze

Parameter	Frozen Value
NG feed	30,000 kg/h
Air feed	45,000 kg/h
Steam feed	87,210 kg/h
Total fresh feed	162,210 kg/h

Parameter	Frozen Value
NH ₃ production	5,453.12 kg/h
NH ₃ production	130.875 t/day
NH ₃ purity	95.4193 mol%
NH ₃ product temperature	30°C
NH ₃ product pressure	145 bar(a)
Heating duty	74.134 MW
Cooling duty	136.381 MW
Internal heat recovery	38.743 MW
Compressor power	0.817 MW

18. Final Design Basis Statement

This Design Basis Document establishes the process design basis for the Greenfield Conventional Ammonia Plant. The design is based on the current DWSIM 10.2 steady-state simulation and supporting project calculations. Detailed equipment calculations, process drawings, control philosophy, relief-system review and process-safety studies shall be developed from this frozen basis. Any subsequent change to feed, product, operating conditions, simulation model or major design assumption shall be reflected in the appropriate project documents and revision-controlled accordingly.